mensik_pset_6

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Problem 1 (a)

$$E[d_i(1)]$$

The expected treatment status i would have when treatment is assigned is 1? Or do we factor in that we know 395 out of the 1,445 total assigned to treatment actually took the treatment, the rest didn't (ITT_D) is approximately 0.273.

```
395/1445
```

```
## [1] 0.2733564
```

$$E[Y_i(0)|d_i(1) = 0]$$

The expected untreated potential outcome for never takers is 36.48

$$E[Y_i(0)|d_i(1) = 1]$$

The expected untreated potential outcome for compliers is 37.54.

$$E[Y_i(1)|d_i(1) = 1]$$

The expected treated potential outcome for compliers is 54.43.

Problem 2

Half compliers, half never takers (not actually treated in 1SNC)

```
schedule <- data.frame(
  Subjects = 1:6,
  Type = c("Complier","Complier","Complier","NT","NT","NT"),
  Yi_0 = c(6, 6, 8, 1, 7, 7),
  Yi_1 = c(3, 1, 2, 7, 7, 30),
  di_1 = c(1,1,1,0,0,0),
  di_0 = c(0,0,0,0,0,0)
)

View(schedule)
```

```r
schedule$tau <- schedule$Yi_1 - schedule$Yi_0
ATE <- mean(schedule$tau)
CACE <- mean(schedule$tau[schedule$type == "Complier"])
```

The ATE is misleading because a subject assigned to treatment that is a never taker skews the ATE positive.

The CACE might be misleading because it does not provide information about never takers. It's possible that if never takers did take the treatment, the effect we are estimating would be different. However, I'm not sure if that is more artifact of the CACE as it is the trouble of never-takers themselves.

Problem 3

3a

If the ITT is negative, the CACE must be negative is true (ITT = ITT_D * CACE)

3b

No - the numerator might not stay constant when the ITTD changes, so we can't assume that the CACE would grow/shrink when ITTD shrinks/grows.

3c

Yes - Y(d=1) effectively does not exist, or can't be observed. The setup is "unidentified."

Problem 4

4a

The expected potential outcomes of subjects when assigned to treatment does equal the expected potential outcome among subjects assigned to treatment, given that they are 'treatment compliers' – these are equal quantities under excludability.

4b

Yes, because both sides of the equation have d_i(1) = 0.

4c

No - these are outcomes of people who don't take the treatment when assigned and people who do take the treatment when assigned. We can't expect the potential outcomes of people who do not take a treatment to be the same as those of people who do take a treatment.

4d

Yes. The meaningful difference in these terms is that they show different assignment status. Excludability lets us only care about potential outcomes vis-a-vis treatments, not treatment assignments.

Problem 5

We can identify compliers: those assigned to treatment who take treatment. That's because in one sided non-comp, we rule out always by assumption.

Among those assigned to control, we can't distinguish who is a complier versus a never taker (both who have di(0)=0) unless a placebo is assigned to the control. It also seems these statuses really depend on the experiment – they are not about what a subject would do in all contexts, and even very similar versions of the same experiment.

Problem 6

6a

```
400/1000*100
```

```
## [1] 40
```

40

```
(700/2000) * 100
```

```
## [1] 35
```

```
40-35
```

```
## [1] 5
```

```
5/(500/1000)
```

```
## [1] 10
```

CACE = 10 percentage points

6b

250 treated – the ITT stays the same, but the ITT_D changes or is now 250/1000 =0.25. The CACE then is 5/0.25, or:

```
5/0.25
```

```
## [1] 20
```

20 percentage points.

6c

If effectiveness is defined as the CACE, then their exaggeration diminishes the reported effect on turnout of their efforts. Given that the ITT is the same whether the true contact rate is reported, it might flummox researchers particularly if they have not themselves canvassed on a particularly hot summer day or in a less than entirely friendly-to-the-cause neighborhood.

Problem 8

```
getwd()
```

```
## [1] "/Users/kom11/Documents/GitHub/regression-study/Experiments"
```

```
q6data <- read.csv("./doi-10.60600-yu-vuh8kd/Rind_Bordia_JASP_1996.csv")
```

a

```
Y1_treated <- (591/1849)*100

Y0_control <- (377/1430)*100

ITT <- Y1_treated - Y0_control

ITT
```

```
## [1] 5.599587
```

Average recycling rates amongst those assigned to treatment were 5.6 percentage points greater than those assigned to control (not canvassed about recycling).

b

```
ITT_D <- 1015 / 1849
ITT_D
```

```
## [1] 0.5489454
```

Among those assigned to treatment, the compliance rate is 54.89.

c

	Assigned treatment	Assigned control
Total	1,849	1,430
Canvassed (d_i)	1,015	0
Not Canvassed	834	1,430
Recycled (Y_i)	591	377
Recycling Rate	591/1849 = 31.96%	337/1430 = 26.36%

Expected recycling rate among those in the control group:

$$E[Y_i(z=0)] = \pi_C E[Y_i(d=0)|C] + (1-\pi_c)E[Y_i(d=0)|N]$$

Expected recycling rate for those assigned to treatment:

$$E[Y_i(z=1)] = \pi_c * E[Y_i(d=1)|C] + (1-\pi_c)E[Y_i(d=0)|N]$$

$$\pi_c = P(d_i(1)=1)$$

$= 0.549$ (the proportion of compliers).

Never-takers treatment receipt does not change across the two expectations. Also, we know that $ITT_Y = E[Y_i(z=1)] - E[Y_i(z=0)]$. We can take out the never takers, subtract the two expected rate equations, and substitute what's left into the above ITTY:

$$ITT_Y = \pi_C(E[Y_i(1)|C] - E[Y_i(0)|C] + (1-\pi_C)(E[Y_i(0)|N] - E[Y_i(0)|N])$$

sans never takers leaves:

$$\pi_C \left(E[Y_i(1) \mid C] - E[Y_i(0) \mid C]\right)$$

which is the CACE times the compliance rate (5.49).

d

```
CACE <- ITT/ITT_D

CACE
```

```
## [1] 10.20063
```

There is a 10.2 percentage point effect of treatment for those who complied with it (10.2 percentage point increase in recycling after being canvassed).

e

Comparing the recycling rates of the treated and untreated subjects alone tends to produce misleading estimates of the CACE because we are not isolating effects of having received treatment.

Problem 10

a

```
getwd()
```

```
## [1] "/Users/kom11/Documents/GitHub/regression-study/Experiments"
```

```r
data_10 <- read.csv("/Users/kom11/Documents/GitHub/regression-study/Experiments/Guan_Green_CPS_2006.csv

data_10 <- na.omit(data_10)

table(data_10$treat2, data_10$contact)
```

```
##
##        0    1
##   0 1334    0
##   1  307 2381
```

```r
ITT <- mean(data_10$turnout[data_10$treat2 == 1] - mean(data_10$turnout[data_10$treat2 == 0]))

ITT
```

```
## [1] 0.1319296
```

The ITT is 0.1319.

b

RI to test sharp null that ITT is zero for all observations, taking into account that random assignment was clustered by dorm room Yi(Z=1) = Yi(Z=0) for each i

```r
# data_10 %>%
#   group_by(dormid) %>%
#   mutate(
#     dorm_pop = n()
#   ) %>%
#   ungroup ()

# I'm confused how to count treated dorms. RI by dorm means that what's supposed to happen is in a dorm

data_10 %>%
  group_by(dormid) %>%
  summarise(
    any_contact = any(contact == 1, na.rm = TRUE)
  ) %>%
  summarise(
    n_dorms_contact = sum(any_contact)
  )
```

```
## # A tibble: 1 x 1
##   n_dorms_contact
##             <int>
## 1             590
```

```r
# 590


treated_dorms <- data_10[data_10$treat2 == 1, ]

length(unique(treated_dorms$dormid))
```

```
## [1] 670
```

```
# 670 dorms assigned to treatment

dorm_check <- data_10 %>%
  group_by(dormid) %>%
  summarise(
    n_assignments = n_distinct(treat2),
    assignment = first(treat2),
    .groups = "drop"
  )

# assignment clustered by dorm room?
all(dorm_check$n_assignments == 1)
```

```
## [1] TRUE
```

```
# n dorms assigned to treatment
n_dorms_assigned <- sum(dorm_check$assignment == 1)

# n dorms rooms in experiment
n_dorms <- nrow(dorm_check)

c(n_dorms = n_dorms, n_dorms_assigned = n_dorms_assigned)
```

```
##          n_dorms n_dorms_assigned
##             1004              670
```

```
# 670 of 1004 dorms assigned

set.seed(6081709)

delcare_10 <- declare_ra(
  N = nrow(data_10),
  clusters = data_10$dormid,
  m = n_dorms_assigned
)

delcare_10
```

```
## Random assignment procedure: Cluster random assignment
## Number of units: 4022
## Number of clusters: 1004
## Number of treatment arms: 2
## The possible treatment categories are 0 and 1.
## The number of possible random assignments is approximately infinite.
## The probabilities of assignment are constant across units:
##    prob_0    prob_1
## 0.3326693 0.6673307
```

```r
permmatrix_10 <- obtain_permutation_matrix(delcare_10)

dim(permmatrix_10)
```

```
## [1]  4022 10000
```

```r
estimates_10 <- numeric(ncol(permmatrix_10))

itt_hat <- mean(data_10$turnout[data_10$treat2 == 1] - mean(data_10$turnout[data_10$treat2 == 0]))

for (i in seq_len(ncol(permmatrix_10))) {

  ri <- permmatrix_10[, i]

  estimates_10 [i] <-
    mean(data_10$turnout[ri == 1], na.rm = TRUE) - mean(data_10$turnout[ri ==0], na.rm = TRUE)
}

mean(estimates_10)
```

```
## [1] -0.0002462874
```

```r
# centered around 0


p_val_10 <- (sum(abs(estimates_10) >= abs(itt_hat)) + 1)/(length(estimates_10)+1) #at first got 0, but

itt_hat
```

```
## [1] 0.1319296
```

```r
p_val_10
```

```
## [1] 9.999e-05
```

The effect of assignment to canvassing was significantly statistic, so we can reject the null hypothesis.

c

CACE assuming the leaflet alone had no effect on turnout seems like the functional equivalent of the exclusion restriction (thing that matters is being treated where treated means canvasser interaction). It is:

```r
itt_hat/(sum(data_10$contact)/sum(data_10$treat2))
```

```
## [1] 0.1489402
```

d

Turnout rate would have been 0.7 instead of 0.8

```r
#turnout for assigned to treatment
2152/2688
```

```
## [1] 0.8005952
```

```r
treated_turnout <- 2152/2688

no_flyer_treated_turnout <- (0.79 * 2688)

no_flyer_treated_turnout
```

```
## [1] 2123.52
```

```r
#assuming the .52 means 1 percentage point is rounded

no_flyer_treated_turnout <- treated_turnout - 0.01

no_flyer_treated_turnout
```

```
## [1] 0.7905952
```

```r
#this was sort of silly to do because I was just told it is 0.8-0.01.

# total proportion compliers
2380/2688
```

```
## [1] 0.8854167
```

$\pi_{d(1)=1} = 2380/2688 \pi_{d(1)=1} = \frac{Y(d=1,z=1)}{Y(z=1)} \mu_{comlier-canvassed-human} = E[Y(z=1,d=1)|d(1)=1] \mu_{complier-not-canvasse}$

Treatment-assigned turnout then:

$$E[Y_i|z_i=1] = \pi_{d(1)=1}\mu_{comlier-canvassed-human} + (1-\pi_{d(1)=1})\mu_{nevertaker-not-canvassed} + 0.01$$

Control-assigned turnout:

$$E[Y_i|z_i=0] = \pi_{d(1)=1}\mu_{complier-not-canvassed} + (1-\pi_{d(1)=1})\mu_{nevertaker-not-canvassed}$$

e

CACE of canvassing

I know E[] treated = 0.7906, and E[] control = 0.6687

```r
prop_complier <- mean(data_10$contact[data_10$treat2 == 1])

# itt hat - 0.01 / pi complier
#
# cace_10e <- itt_hat - 0.01 /pi_complier

control_turnout <- mean(data_10$turnout[data_10$treat2 == 0])

#treated_turnout
#no_flyer_treated_turnout

cace_canvassing <- (
  no_flyer_treated_turnout - control_turnout
) / prop_complier

cace_canvassing
```

```
## [1] 0.1376509
```

The CACE is 0.137

f

CACE if leaflet had no effect on compliers and raised turnout among never takers by 3 pct points. Have: ITT_hat proportion compliers (prop_complier) Y_hat for control (control_turnout) Y-hat for treated when all assigned to treatment are counted as treated (treated_turnout)

```r
#change to ITT is 0.03 * prop never takers
# is 0.03 * (1-prop_complier)

prop_nts <- 1-prop_complier

new_itt <- itt_hat-(.03*prop_nts)

# cace is new / prop_complier

new_itt / prop_complier
```

```
## [1] 0.1450721
```